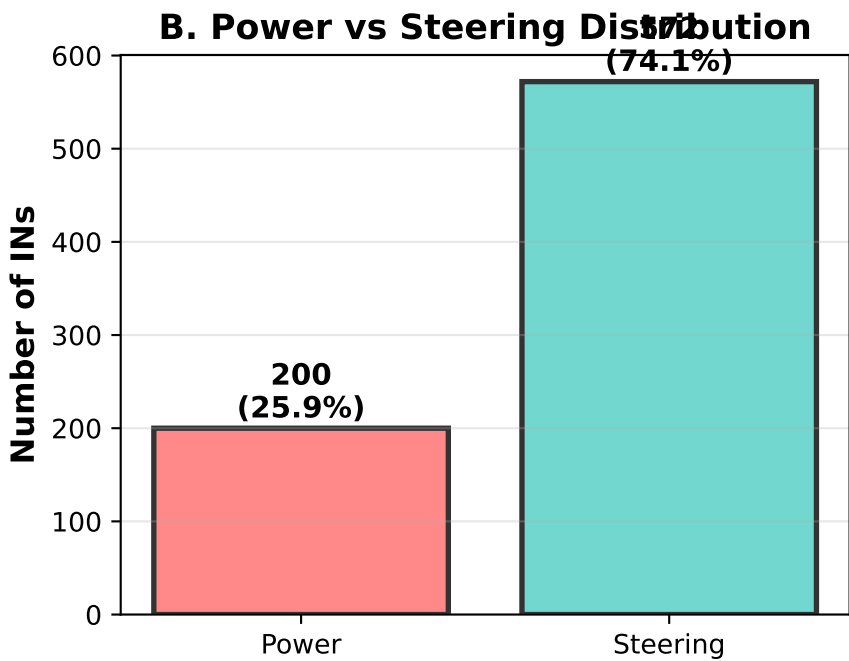
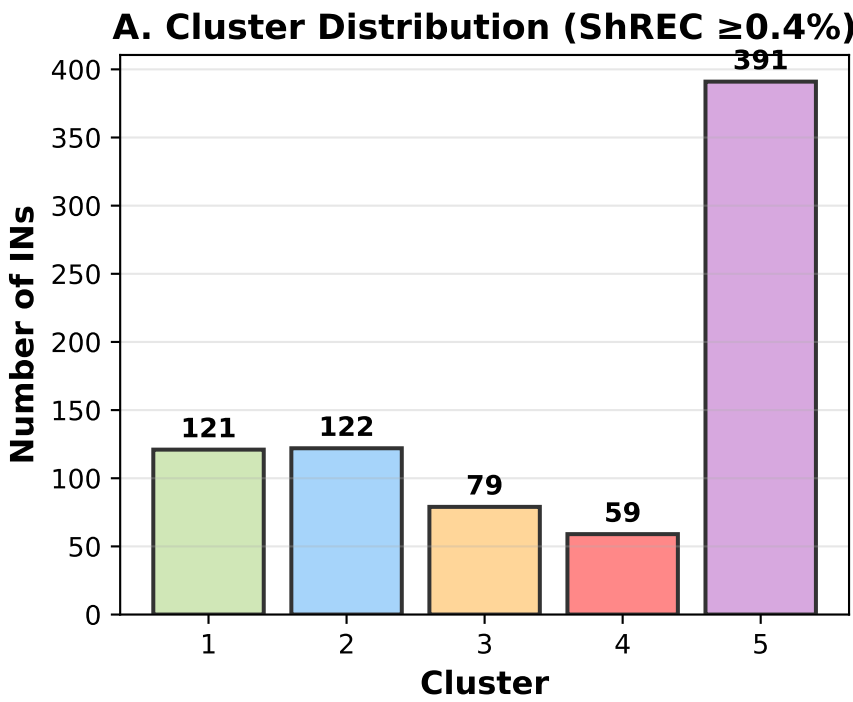


Complete ShREC-Based Flight Control Circuit Analysis (Threshold: 0.4%)



CLUSTERING METRICS:

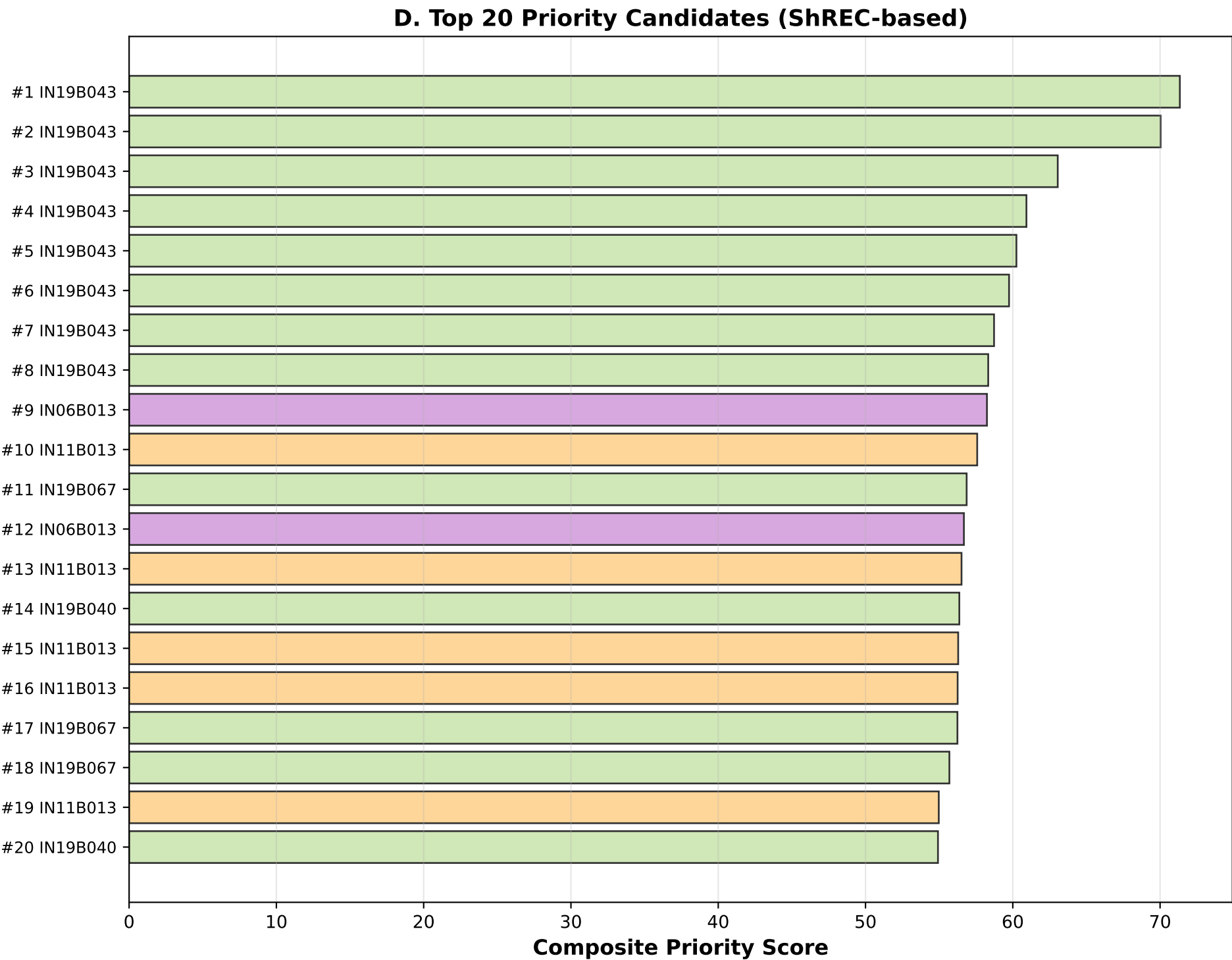
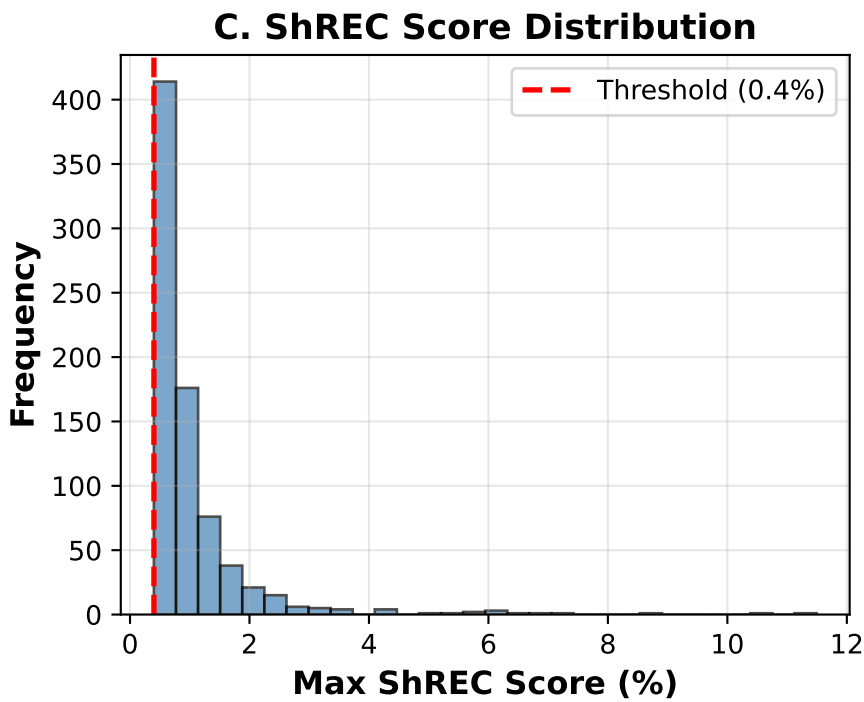
Silhouette Score: 0.4197
(Range: -1 to 1)
(Yours: Good)

Calinski-Harabasz: 69.8
(Higher = Better)

Number of Clusters: 5

- 1 Power control
- 4 Steering control

Total Premotor INs: 772
Hub Neurons: 443



ShREC-BASED ANALYSIS SUMMARY

METHODOLOGY:

- Functional connectivity threshold: ShREC $\geq 0.4\%$
- Clustering: Hierarchical (Ward's method, cosine similarity)
- Hub identification: Network centrality (median threshold)

NETWORK STATISTICS:

- Total premotor INs: 772
- Hub neurons: 443 (57.4%)
- Functional modules: 5 (validated by modularity)

POWER CONTROL:

- INs: 200 (25.9%)
- Avg syn/IN: 121.1
- Strategy: Robust, reliable control

STEERING CONTROL:

- INs: 572 (74.1%)
- Avg syn/IN: 53.2
- Strategy: Distributed, flexible control

TOP PRIORITY NEURON:

- Rank: #1
- Name: IN19B043
- Function: power_control
- Hub status: True
- Max ShREC: 6.20%
- MN targets: 34

ADVANTAGES OF ShREC APPROACH:

- ✓ Accounts for relative functional impact
- ✓ Comparable to Ache et al. 2025 methodology
- ✓ Filters biologically meaningful connections
- ✓ Independent of absolute synapse counts